=== UNIT 3: INTEGRAL CALCULUS, REDUCTION FORMULAE & BETA-GAMMA FUNCTIONS ===

>> 3.1 Definite Integrals and Reduction Formulae

Integral calculus underpins the computation of continuous mass balances, cumulative reaction kinetics, and electromagnetic potentials.

* Reduction Formula for sin^n(x): Derived using integration by parts (u = sin^{n-1}(x), dv = sin(x) dx):

I_n = int sin^n(x) dx = - [sin^{n-1}(x)*cos(x) / n] + [(n - 1) / n] * I_{n-2}.

* Reduction Formula for cos^n(x):

I_n = int cos^n(x) dx = [cos^{n-1}(x)*sin(x) / n] + [(n - 1) / n] * I_{n-2}.

* Walli's Formula for Definite Integrals over [0, pi/2]:

int_0^{pi/2} sin^n(x) dx = int_0^{pi/2} cos^n(x) dx = [(n-1)*(n-3)*...*(1 or 2)] / [n*(n-2)*...*(2 or 1)] * K,

where K = pi/2 if n is an EVEN positive integer, and K = 1 if n is an ODD positive integer.

* Walli's Product Formula for int_0^{pi/2} sin^p(x) * cos^q(x) dx:

= [(p-1)*(p-3)... * (q-1)*(q-3)...] / [(p+q)*(p+q-2)*(p+q-4)...] * K,

where K = pi/2 if BOTH p and q are even integers, and K = 1 otherwise.

>> 3.2 Special Functions: Beta and Gamma Functions

* The Gamma Function (Euler's Integral of Second Kind): Defined for all real x > 0 by:

Gamma(n) = int_0^inf e^{-t} * t^{n - 1} dt.

- Fundamental Recurrence Relation: Gamma(n + 1) = n * Gamma(n).
- Factorial Representation: For positive integer n, Gamma(n + 1) = n! and Gamma(n) = (n - 1)!.
- Important Exact Value: Gamma(1/2) = sqrt(pi).
- Negative Non-integers: Evaluated via Gamma(n) = Gamma(n + 1) / n.

* The Beta Function (Euler's Integral of First Kind): Defined for m > 0, n > 0 by:

Beta(m, n) = int_0^1 x^{m - 1} * (1 - x)^{n - 1} dx.

- Trigonometric Form: Beta(m, n) = 2 * int_0^{pi/2} sin^{2m - 1}(theta) * cos^{2n - 1}(theta) d(theta).
- Symmetry Property: Beta(m, n) = Beta(n, m).
- Algebraic Transformation: Beta(m, n) = int_0^inf [y^{m - 1} / (1 + y)^{m + n}] dy.

* Fundamental Relation Between Beta and Gamma Functions:

Beta(m, n) = [Gamma(m) * Gamma(n)] / Gamma(m + n).

- Duplication Formula (Legendre): Gamma(m) * Gamma(m + 1/2) = [sqrt(pi) / 2^{2m - 1}] * Gamma(2m).

>> 3.3 Rectification and Area Computation of Plane Curves

* Arc Length (Rectification) in Cartesian Coordinates: s = int_a^b sqrt(1 + (dy/dx)^2) dx.
* Arc Length in Parametric Coordinates: s = int_{t1}^{t2} sqrt((dx/dt)^2 + (dy/dt)^2) dt.
* Arc Length in Polar Coordinates: s = int_{theta1}^{theta2} sqrt(r^2 + (dr/d(theta))^2) d(theta).

[SOLVED PROBLEM 1: Application of Walli's Definite Integral Formula]

Problem: Evaluate the definite integral I = int_0^{pi/2} sin^6(x) * cos^4(x) dx.
Solution Step 1: Identify indices p and q:
p = 6 (even positive integer), q = 4 (even positive integer).
Since BOTH p and q are even, the multiplier factor K = pi/2.
Solution Step 2: Apply Walli's formula:
Numerator terms: (p - 1)*(p - 3)*(p - 5) * (q - 1)*(q - 3) = (5 * 3 * 1) * (3 * 1) = 15 * 3 = 45.
Denominator terms: (p + q)*(p + q - 2)*(p + q - 4)... down to 2:
p + q = 10.
Denominator = 10 * 8 * 6 * 4 * 2 = 3,840.
Solution Step 3: Multiply by K = pi/2:
I = [45 / 3840] * (pi / 2) = [3 / 256] * (pi / 2) = 3*pi / 512.
Alternative Solution via Beta-Gamma Relation:
int_0^{pi/2} sin^6(x) * cos^4(x) dx = (1/2) * Beta((6+1)/2, (4+1)/2) = (1/2) * Beta(7/2, 5/2)
= (1/2) * [Gamma(7/2) * Gamma(5/2)] / Gamma(6).
Gamma(7/2) = (5/2)*(3/2)*(1/2)*sqrt(pi) = (15/8)*sqrt(pi).
Gamma(5/2) = (3/2)*(1/2)*sqrt(pi) = (3/4)*sqrt(pi).
Gamma(6) = 5! = 120.
I = (1/2) * [(15/8 * 3/4 * pi) / 120] = (1/2) * [(45/32 * pi) / 120] = (1/2) * (3*pi / 256) = 3*pi / 512. (Confirmed).

[SOLVED PROBLEM 2: Evaluation of Non-Elementary Gaussian Integral via Gamma]

Problem: Evaluate the definite integral J = int_0^inf x^4 * e^{-2x^2} dx using the Gamma function.
Solution Step 1: Variable substitution to match standard Gamma integral int_0^inf t^{n-1} e^{-t} dt:
Let t = 2*x^2 => x^2 = t / 2 => x = (t / 2)^{1/2}.
Differentiating: dt = 4*x dx => dx = dt / (4*x) = dt / (4 * (t/2)^{1/2}) = (1 / (2*sqrt(2))) * t^{-1/2} dt.

Solution Step 2: Transform limits:
When x = 0, t = 0; when x -> inf, t -> inf.
Solution Step 3: Substitute into integral:
x^4 = (x^2)^2 = (t / 2)^2 = t^2 / 4.
J = int_0^inf [t^2 / 4] * e^{-t} * [(1 / (2*sqrt(2))) * t^{-1/2} dt]
J = [1 / (8*sqrt(2))] * int_0^inf t^{2 - 1/2} * e^{-t} dt
J = [1 / (8*sqrt(2))] * int_0^inf t^{3/2} * e^{-t} dt.
Solution Step 4: Recognize Gamma integral with n - 1 = 3/2 => n = 5/2:
int_0^inf t^{3/2} * e^{-t} dt = Gamma(5/2).
Gamma(5/2) = (3/2) * (1/2) * Gamma(1/2) = (3/4) * sqrt(pi).
Solution Step 5: Final computation:
J = [1 / (8*sqrt(2))] * [(3/4) * sqrt(pi)] = 3 * sqrt(pi) / (32 * sqrt(2)) = [3 *
sqrt(2*pi)] / 64.
Final Numerical Value: J ~ 3 * 2.5066 / 64 = 0.1175.

[SOLVED PROBLEM 3: Proof of Beta-Gamma Relation Beta(m,n) = Gamma(m)*Gamma(n)/Gamma(m+n)]

Problem: Rigorously prove the identity Beta(m, n) = [Gamma(m) * Gamma(n)] / Gamma(m +
n).
Proof Step 1: Express Gamma functions using substitute variables:
Gamma(m) = int_0^inf e^{-x^2} * (x^2)^{m - 1} * (2x dx) = 2 * int_0^inf e^{-x^2} * x^{2m
- 1} dx.
Gamma(n) = 2 * int_0^inf e^{-y^2} * y^{2n - 1} dy.
Proof Step 2: Multiply the two integrals as a double integral over the first quadrant (x
>= 0, y >= 0):
Gamma(m) * Gamma(n) = 4 * int_0^inf int_0^inf e^{-(x^2 + y^2)} * x^{2m - 1} * y^{2n - 1}
dx dy.
Proof Step 3: Transform to polar coordinates (x = r*cos(theta), y = r*sin(theta), dx dy
= r dr d(theta)):
Limits: r runs from 0 to inf; theta runs from 0 to pi/2.
x^2 + y^2 = r^2; x^{2m - 1} = r^{2m - 1} * cos^{2m - 1}(theta); y^{2n - 1} = r^{2n - 1}
* sin^{2n - 1}(theta).
Gamma(m)*Gamma(n) = 4 * int_0^{pi/2} sin^{2n-1}(theta)*cos^{2m-1}(theta) d(theta) *
int_0^inf e^{-r^2} * r^{2m+2n-2} * r dr.
Proof Step 4: Recognize separated integral components:
First integral: 2 * int_0^{pi/2} sin^{2n-1}(theta) * cos^{2m-1}(theta) d(theta) =
Beta(n, m) = Beta(m, n).
Second integral: 2 * int_0^inf e^{-r^2} * r^{2(m+n) - 1} dr = Gamma(m + n).
Therefore: Gamma(m) * Gamma(n) = Beta(m, n) * Gamma(m + n).
Dividing by Gamma(m + n): Beta(m, n) = [Gamma(m) * Gamma(n)] / Gamma(m + n). (Q.E.D.).

[SOLVED PROBLEM 4: Arc Length (Rectification) of Cardioid]

Problem: Find the total perimeter (arc length) of the complete cardioid curve r = a * (1 + cos(theta)).
Solution Step 1: Utilize symmetry: Cardioid is symmetric about the initial polar axis (theta = 0).
Total length S = 2 * int_0^pi sqrt(r^2 + (dr/d(theta))^2) d(theta).
Solution Step 2: Calculate derivative and differential element:
dr/d(theta) = -a * sin(theta).
r^2 + (dr/d(theta))^2 = a^2 * (1 + cos(theta))^2 + a^2 * sin^2(theta)
= a^2 * [1 + 2*cos(theta) + cos^2(theta) + sin^2(theta)] = a^2 * [2 + 2*cos(theta)]
= 2*a^2 * [1 + cos(theta)] = 2*a^2 * [2 * cos^2(theta / 2)] = 4*a^2 * cos^2(theta / 2).
Solution Step 3: Take square root (for 0 <= theta <= pi, cos(theta/2) >= 0):
sqrt(r^2 + (dr/d(theta))^2) = 2*a * cos(theta / 2).
Solution Step 4: Integrate from 0 to pi:
S = 2 * int_0^pi [2*a * cos(theta / 2)] d(theta) = 4*a * [2 * sin(theta / 2)]_0^pi
S = 8*a * [sin(pi / 2) - sin(0)] = 8*a * (1 - 0) = 8*a.
Final Answer: The total perimeter of the cardioid is 8a.

>> 3.4 University Review & Examination Questions

* Q1: Evaluate int_0^1 [x^2 / sqrt(1 - x^4)] dx using Beta and Gamma functions.
Answer: Put x^2 = sin(theta) => 2x dx = cos(theta) d(theta) => dx = cos(theta) d(theta) / (2*sqrt(sin(theta))). As x: 0->1, theta: 0->pi/2. The integral becomes int_0^{pi/2} [sin(theta) / cos(theta)] * [cos(theta) / (2*sqrt(sin(theta)))] d(theta) = (1/2) * int_0^{pi/2} sin^{1/2}(theta) * cos^0(theta) d(theta). Using the Beta relation: = (1/4) * Beta((1/2 + 1)/2, (0 + 1)/2) = (1/4) * Beta(3/4, 1/2) = (1/4) * [Gamma(3/4) * Gamma(1/2)] / Gamma(5/4). Since Gamma(5/4) = (1/4)*Gamma(1/4), this simplifies to [Gamma(3/4) * sqrt(pi)] / Gamma(1/4).
* Q2: Derive the reduction formula for int tan^n(x) dx and hence evaluate int_0^{pi/4} tan^5(x) dx.
Answer: Write tan^n(x) = tan^{n-2}(x) * tan^2(x) = tan^{n-2}(x) * (sec^2(x) - 1) = tan^{n-2}(x)*sec^2(x) - tan^{n-2}(x). Integrating: I_n = [tan^{n-1}(x) / (n - 1)] - I_{n-2}. For limits 0 to pi/4, tan(pi/4) = 1 and tan(0) = 0, so I_n = 1/(n-1) - I_{n-2}.
For I_5: I_5 = 1/4 - I_3; I_3 = 1/2 - I_1; I_1 = int_0^{pi/4} tan(x) dx = [ln(sec(x))]_0^{pi/4} = ln(sqrt(2)) = (1/2)*ln(2). Thus I_3 = 1/2 - (1/2)*ln(2), and I_5 = 1/4 - [1/2 - (1/2)*ln(2)] = (1/2)*ln(2) - 1/4.

>> 3.5 Key Takeaways and Revision Points

* Walli’s formula directly evaluates int_0^{pi/2} sin^p(x)*cos^q(x) dx using parity of indices.
* Gamma function generalizes factorial: Gamma(n+1) = n*Gamma(n); Gamma(1/2) = sqrt(pi).

* Beta-Gamma identity Beta(m,n) = Gamma(m)*Gamma(n) / Gamma(m+n) converts complex non-elementary integrals into products of Gamma functions.
* Rectification computes exact arc lengths using differential metric ds = sqrt(1 + (y')^2) dx.